

Decoding Customer Value

A data-driven retention & acquisition strategy for a D2C fashion brand

IIT Guwahati Consulting & Analytics Club · 3,900-customer transactional dataset · Python feature engineering → SQL segmentation → Power BI

The problem. The brand was spending heavily on blanket discounts and broad-age acquisition without knowing which customers actually drive margin. We engineered a customer-value model (CLV from spend × tenure, plus two independent loyalty definitions), used SQL to split the base into **five behavioral segments**, and surfaced the results in a Power BI decision dashboard. The headline: a large share of discount spend is going to customers who would very likely pay full price anyway.

3,900

CUSTOMERS ANALYZED

5

VALUE SEGMENTS

424

PRIORITY SUNSET
SEGMENT

~\$1.1M

ANNUAL SPEND
EXPOSED

**\$130-
220K**

EST. MARGIN RECOVERY

Key Findings

1 Discounts are subsidizing already-loyal value

Promo Reliant–Premium (424) and Champions (551) are near-identical on CLV (~\$8,500) and annual spend (~\$2,600) — yet Champions buy at **0% discount dependence** vs **100%** for Promo Reliant. Same value, opposite reliance on promos.

2 Behavior predicts value; demographics don't

Purchase frequency separates tiers sharply (**Premium 38×/yr vs Low 2.4×/yr**), while average age is ~44 across every tier. Targeting by age alone wastes spend; frequency and tenure are the real signals.

3 Specific geographies show genuine brand pull

Tennessee leads on willingness to pay full price (high spend, just 36.4% promo dependency), followed by Illinois, Alaska and Pennsylvania. Indiana's revenue is discount-fragile (57% promo dependency).

4 The ideal customer is narrow and identifiable

Highest-value, full-price loyalists are women buying **Clothing** on Credit Card / PayPal — **26–35 in Fall** (~\$3,575 spend) and **46–55 in Winter** (~\$3,484). The value-draining anti-profile: 18–25 buying Accessories with cash on promo.

Recommendations

1 Launch a phased Promotional Sunset

Wean Promo Reliant–Premium off discounts — the safest high-value test group (~\$1.1M exposed). Step down over **9 months (25% → 50% → full removal)**, with go/no-go gates at **≥85% retention** and **≤10% spend drop**. Breakeven ~12% churn. **Pilot in Tennessee first.**

2 Reallocate ~40% of acquisition budget

Shift top-of-funnel social spend from broad 18–35 targeting to the high-value female Clothing ICP (46–55 Winter, 26–35 Spring/Fall) in **TN, IL, PA**. Lead with quality and exclusivity, not discount codes.

3 Deprioritize the anti-profile

Apply negative targeting to 18–25 / Accessories / cash-and-promo buyers, and drop Accessories as a paid entry category. Accept lower vanity conversion in exchange for higher margin per acquired customer.

4 Gate launches behind loyalty, not discounts

Release best collections through early-access and loyalty sign-ups rather than sitewide discounts — converting impulse, promo-driven demand into full-price, high-CLV advocates.

BOTTOM LINE

The growth lever is not more discounts or more traffic — it is **reallocating spend toward customers who already pay full price, and acquiring more like them**. The segmentation shows this can be executed with controlled, measurable risk: start with one high-value segment, in one high-pull market, behind clear retention guardrails.

Method note: $CLV_Score = (Purchase\ Amount \times annualized\ frequency) \times \log(1 + previous\ purchases)$. Segments derived from CLV value-tier quartiles and promo-dependence. Loyalty defined two ways — A: frequency + tenure + promo-independence; B: CLV + satisfaction + promo-independence — with "genuinely loyal" requiring both. The 15–20% discount rate behind the margin estimate is an assumption (dataset records only whether a promo was used); recompute once the true average discount is known.